

Feature Detection: Edges

Differential Detectors/Operators

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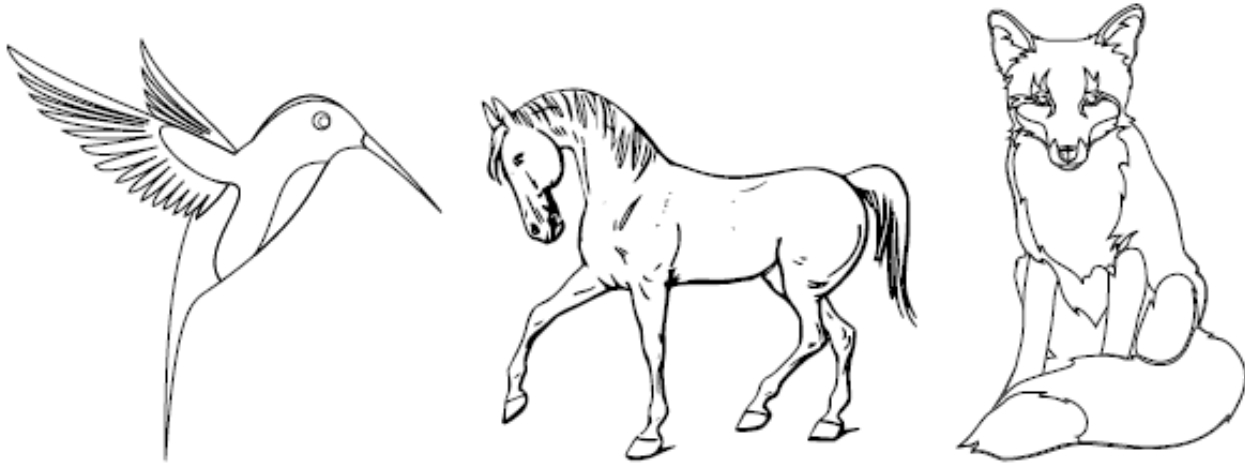
- Image thresholding
 - Is a simple way to achieve image segmentation
 - Can be used in binary segmentation, i.e. it sets a label for pixels with intensities lower than a given value (called threshold level) and the remaining pixels with another label
 - Produces great results for image obtained in all kinds of environments
- K-means
 - Is a clustering algorithm
 - Is fully automatic and does not need parameter setting
 - Can cluster using any number of features (>1), however, as the number of features grow, it will become very slow
 - Will find the number of clusters in a feature space

Outline

- Importance of Edges
- Image Derivatives
- Image Gradient
- Second order derivatives

Edges Are Important

- We can understand what's what just by the edges
- Our visual system gives a lot of importance to edges



Causes of edges

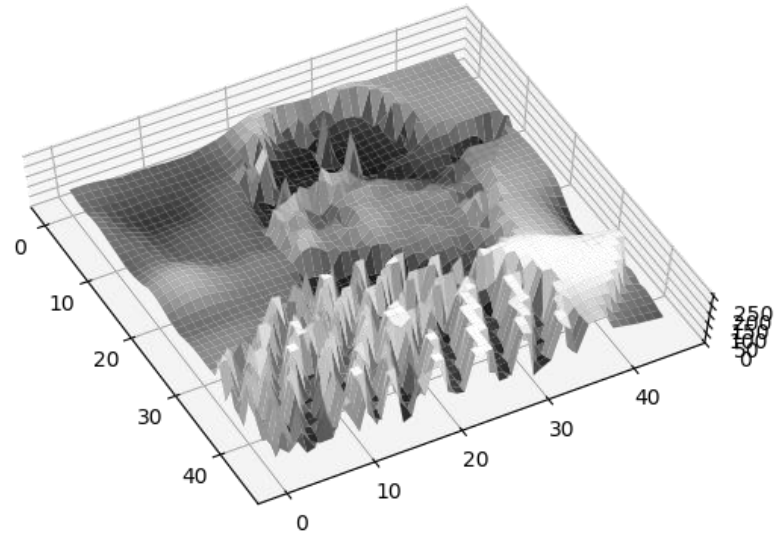
- Discontinuity in:
 - Surface normal
 - Surface color
 - Depth
 - Illumination
- Important
 - Feature extraction
 - Object recognition



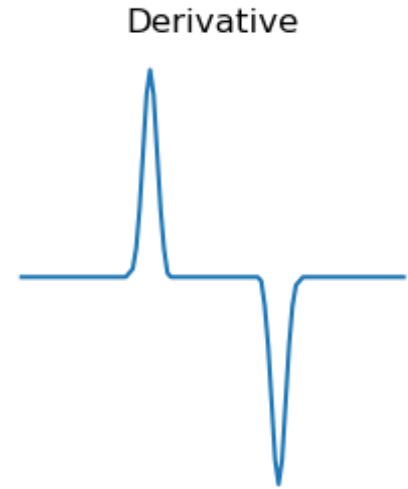
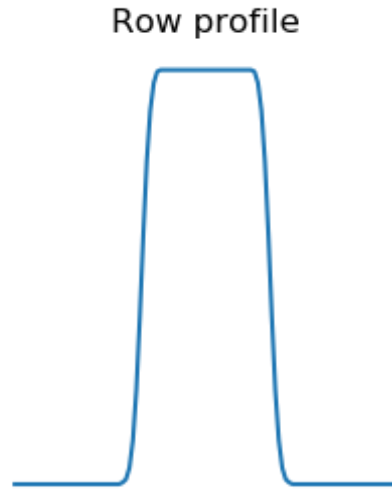
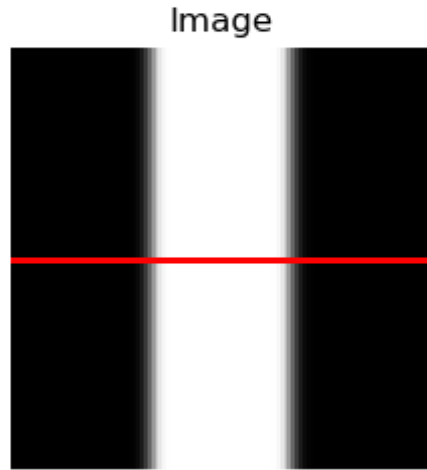
Criteria for optimal edge detection

- Detection
 - Minimize misses (false negatives)
 - Minimize false positives
- Localization
 - Detected edge should be as close as possible to the real edge

Back to “image as a function”



First order derivatives



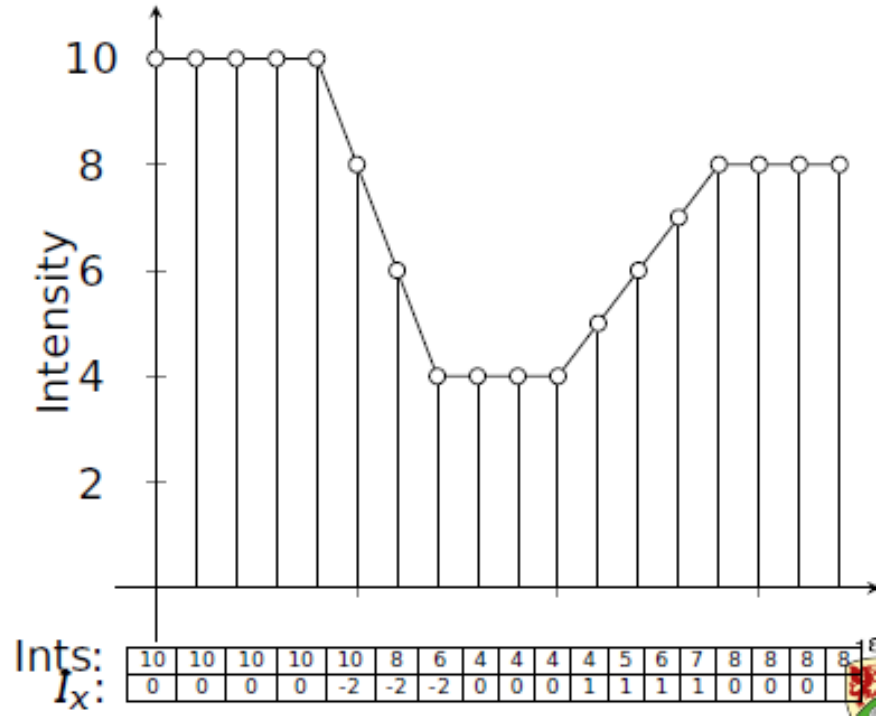
Understanding the derivative

- Derivative:

$$\frac{d}{dx}f = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- Approximation:

$$\frac{d}{dx}I \approx I_x = I(x+1) - I(x)$$



Discrete derivatives (1D)

- Forward:

$$\frac{d}{dx}I \approx I(x+1) - I(x)$$

- Backward:

$$\frac{d}{dx}I \approx I(x) - I(x-1)$$

- Central:

$$\frac{d}{dx}I \approx \frac{I(x+1) - I(x-1)}{2}$$

Discrete derivatives (2D)

- For functions with more than one variable, we need partial

$$\nabla I = \begin{bmatrix} \frac{\partial}{\partial x} I \\ \frac{\partial}{\partial y} I \end{bmatrix}$$

$$\frac{\partial}{\partial x} I = \frac{I(x+1) - I(x-1)}{2}, \quad \frac{\partial}{\partial y} I = \frac{I(y+1) - I(y-1)}{2}$$

Prewitt operator

- Effect of the noise is reduced by taking the contribution of neighbours
- Almost like an weighted average derivative

$$\frac{\partial}{\partial x}I = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} * I$$

$$\frac{\partial}{\partial y}I = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} * I$$

- Note that:

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

Sobel operator

- Effect of the noise is reduced by taking the contribution of neighbours
- Almost like an weighted average derivative (Gaussian of derivative)

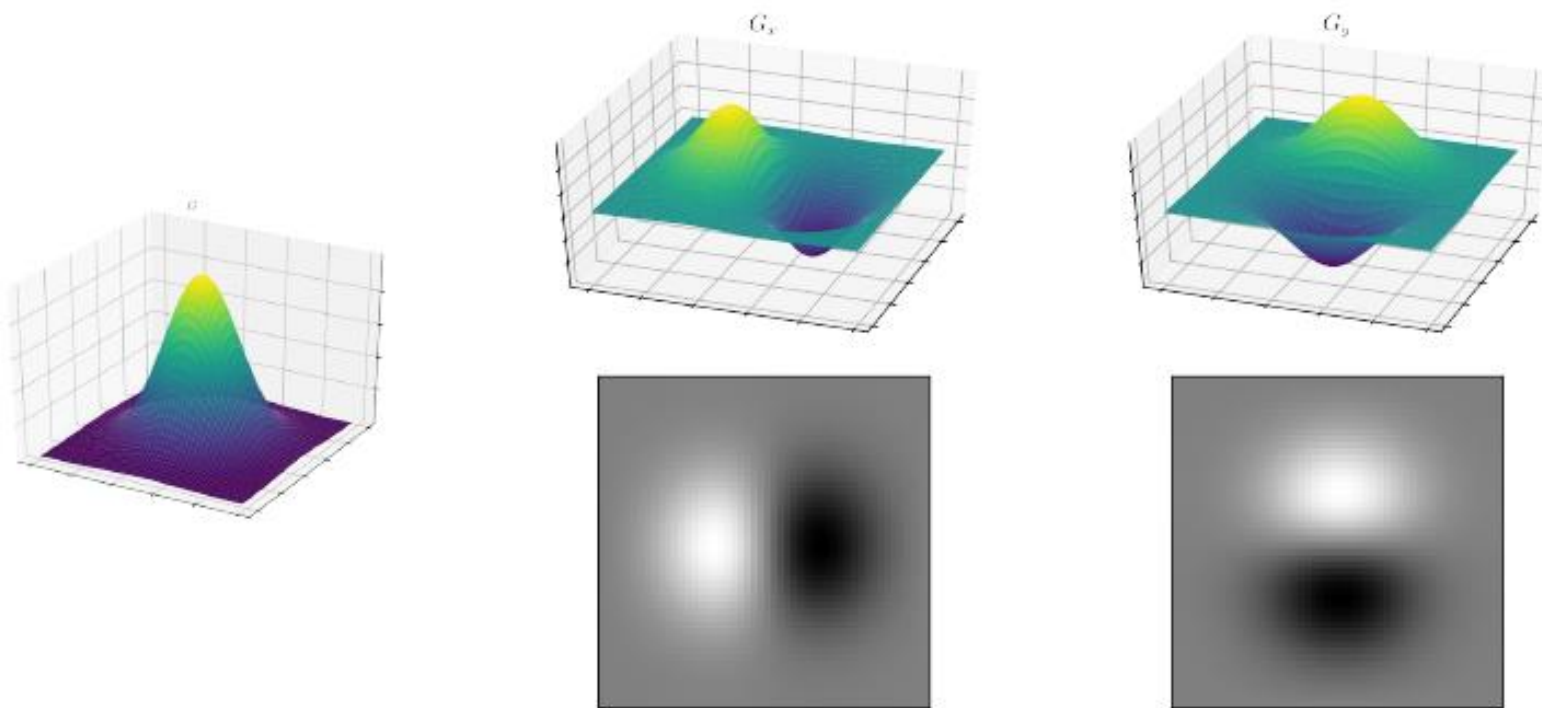
$$\frac{\partial}{\partial x}I = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} * I$$

$$\frac{\partial}{\partial y}I = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} * I$$

- Note that:

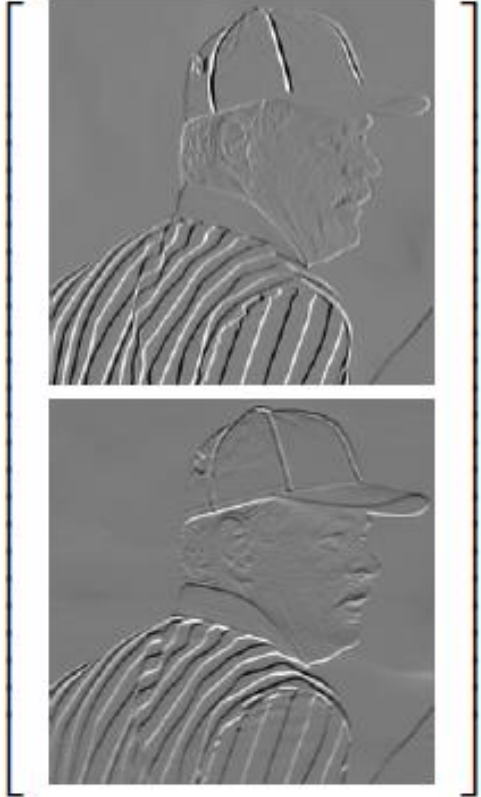
$$\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Sobel $\approx$ Derivative of Gaussian



“Visualizing” the gradient

$$\nabla I = \begin{bmatrix} \frac{\partial}{\partial x} I \\ \frac{\partial}{\partial y} I \end{bmatrix} =$$



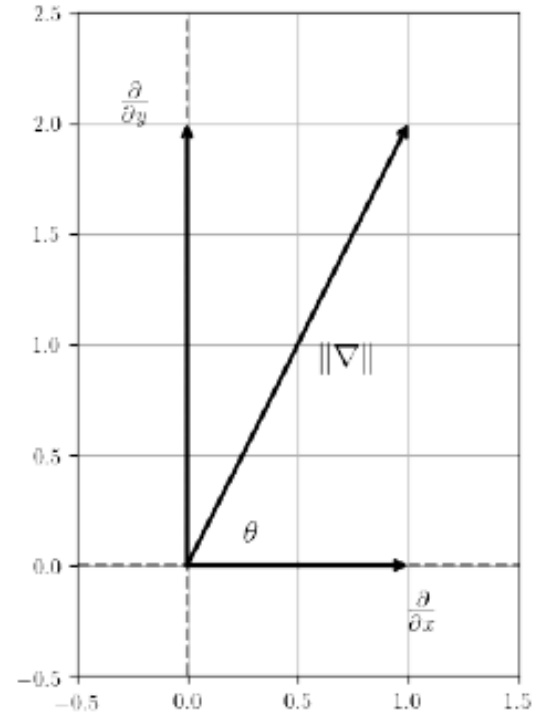
Gradient is a vector

- Vectors have
 - Magnitude:

$$\|\nabla I\| = \sqrt{\left(\frac{\partial}{\partial x} I\right)^2 + \left(\frac{\partial}{\partial y} I\right)^2}$$

- Angle of the direction:

$$\theta = \arctan\left(\frac{\frac{\partial}{\partial y} I}{\frac{\partial}{\partial x} I}\right)$$



“Visualizing” the magnitude and direction

$$\|\nabla I\| =$$



$$\arctan\left(\frac{I_y}{I_x}\right) =$$

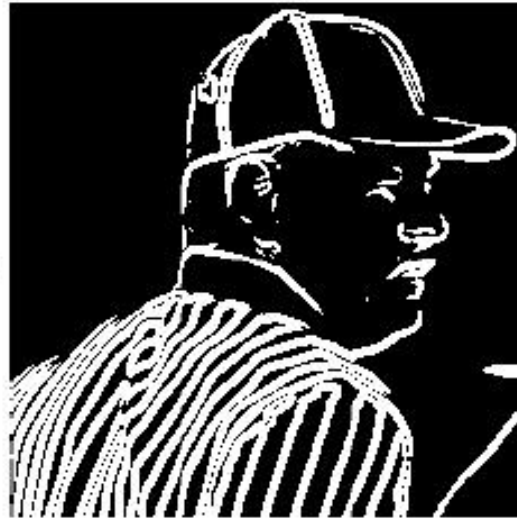


Edges from thresholded derivative

$$\|\nabla I\| =$$



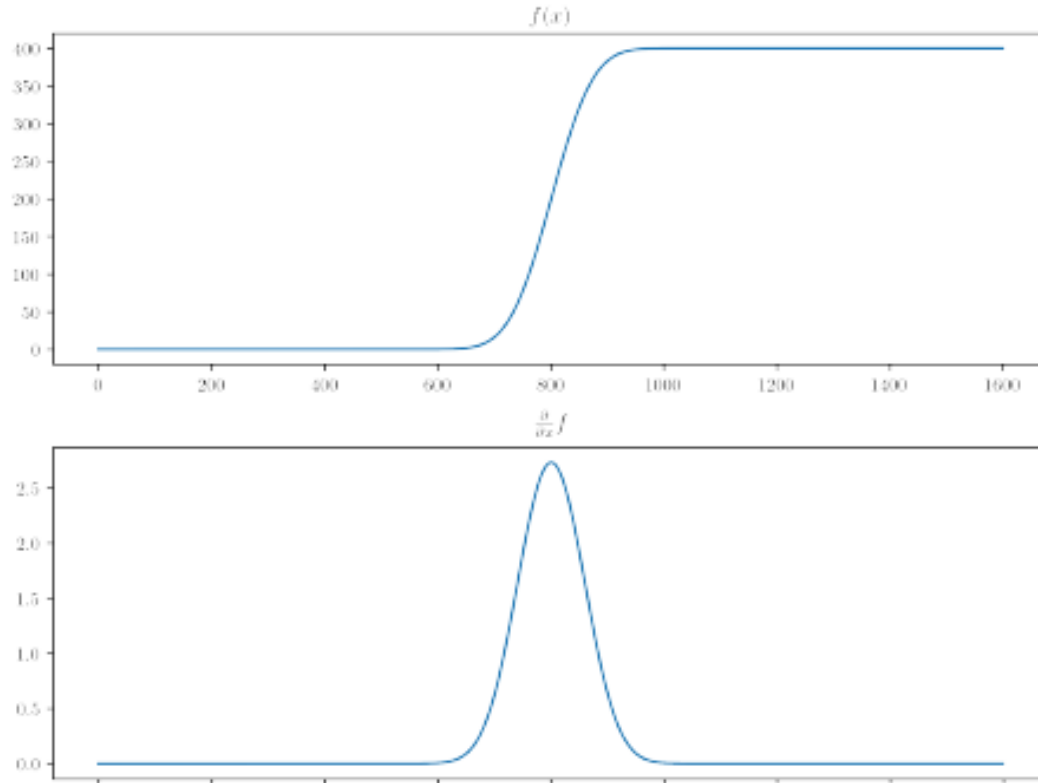
$$\|\nabla I\| \geq T =$$



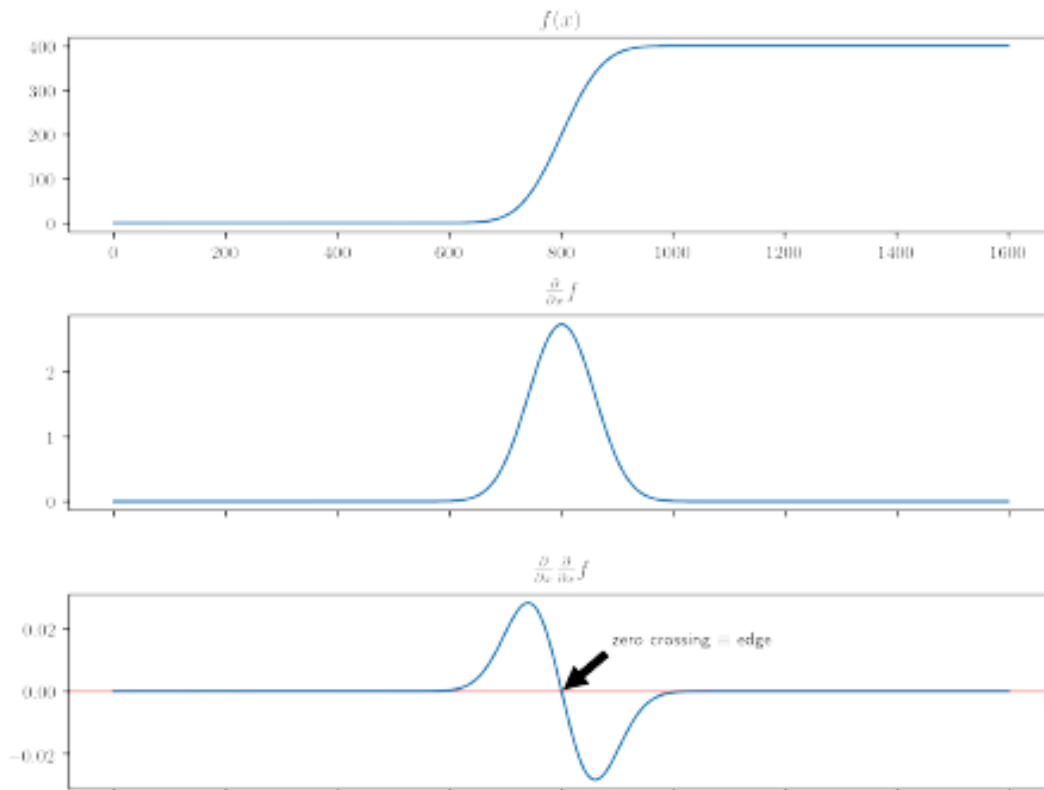
Another look at the direction matrix



Edge detection: Assuming no noise



Edge as zero-crossing of 2nd derivative



Second order derivatives

- Second order differential operator (derivative of a derivative)

$$\frac{\partial^2}{\partial x^2} f = \frac{d}{dx} \left(\frac{d}{dx} f \right)$$

- Discrete 1D approximation:

$$\frac{\partial^2}{\partial x^2} f \approx f(x+1) - 2f(x) + f(x-1)$$

Laplacian 2D

- When we have 2 dimensions, the Laplacian is defined as:

$$\Delta I = \frac{\partial^2}{\partial x^2} I + \frac{\partial^2}{\partial y^2} I$$

- The discrete kernel approximation is:

$$\Delta I = \begin{array}{|c|c|c|} \hline 0 & 0 & 0 \\ \hline 1 & -2 & 1 \\ \hline 0 & 0 & 0 \\ \hline \end{array} + \begin{array}{|c|c|c|} \hline 0 & 1 & 0 \\ \hline 0 & -2 & 0 \\ \hline 0 & 1 & 0 \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline 0 & 1 & 0 \\ \hline 1 & -4 & 1 \\ \hline 0 & 1 & 0 \\ \hline \end{array}$$

Looking at the Laplacian

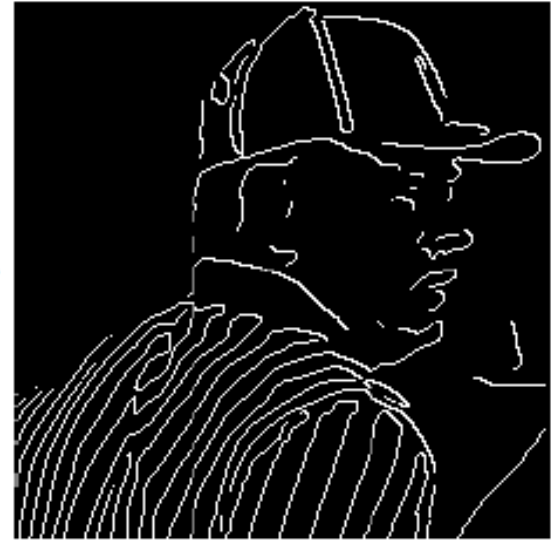


Thresholded deri vs. zero-crossing of 2nd deri

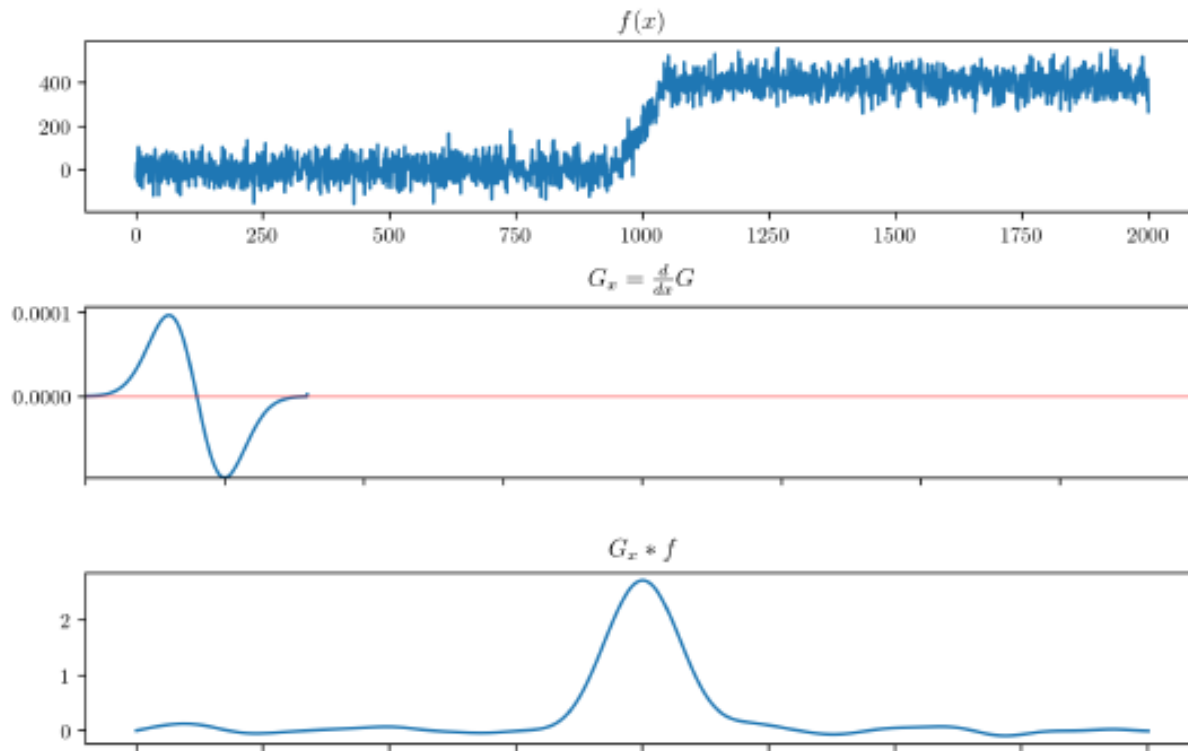
$$\|\nabla I\| \geq T \rightarrow$$



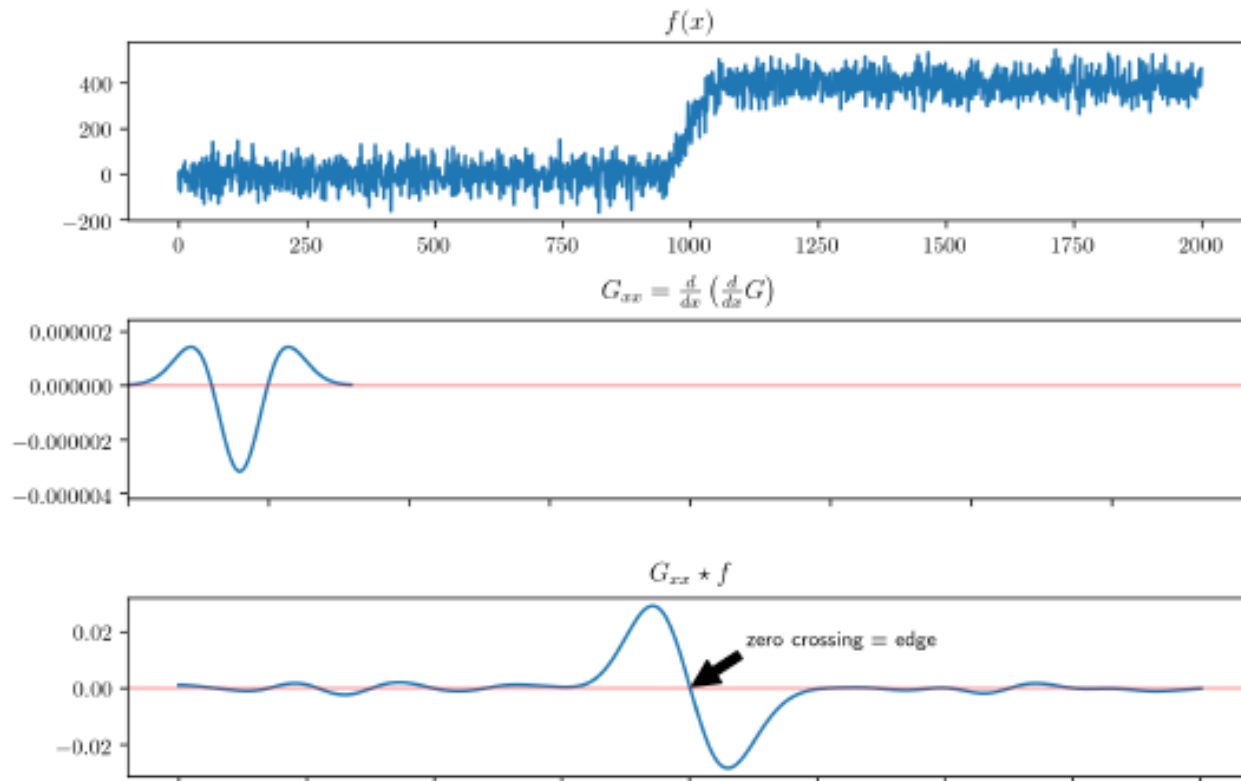
$$\Delta I \approx 0 \rightarrow$$



“Real-world” convolution



Where is the edge



Summary

- Differential operators are very sensitive to noise
- The higher the order the more sensitive
- Edges can be “extracted” from:
 - Local maxima (or minima) of first derivative
 - Acceptable results if no noise and sharp edges
 - Otherwise, produces wide edges (non-localized)
 - Zero-crossings of second derivative (e.g. LoG)
 - Better localization